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Intranasal administration of the essential oil from Perillae Folium ameliorates social defeat stress-induced behavioral impairments in mice

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ABSTRACT

Ethnopharmacological relevance: Perillae Folium, the leaves and twigs of *Perilla frutescens* (L.) Britton, has been included in many traditional Chinese medicine herbal formulas to treat depression. However, the precise anti-depressant mechanism of the essential oil from Perillae Folium (PFEO) has not been fully investigated.

Aim of the study: To assess the effects and potential mechanisms of PFEO on depression using animal models and network pharmacology analysis.

Materials and methods: PFEO was intranasally administered to a mouse model of social defeat stress (SDS). The antidepressant effects of PFEO on SDS-induced mice were evaluated using behavioral tests. Enzyme-linked immunosorbent assay (ELISA) and western blot were performed to measure the levels of depression-related biomarkers in the hippocampus and serum of the mice. The chemical compounds of PFEO were determined using gas chromatography-mass spectrometry (GC-MS). Network pharmacology and molecular docking analyses were conducted to investigate the potential bioactive components of PFEO and the mechanisms underlying the antidepressant effects. To validate the mechanisms of the bioactive compounds, *in vitro* models using PC12 and BV2 cells were established and the blood-brain barrier (BBB) permeability was evaluated.

Results: The intranasal administration of PFEO suppressed SDS-induced depression in mice by increasing the time spent in the social zone and the social interactions in the social interaction test and by decreasing the immobility time in the tail suspension and forced swimming tests. Moreover, the PFEO treatment reduced the SDS-induced anxiety-like behavior, as inferred from the increased activity in the central zone observed in the open field test and in the open arms observed in the elevated plus maze test. PFEO administration recovered the SDS-induced decrease in the levels of 5-HT, NE, gamma-aminobutyric acid (GABA), and p-ERK in the hippocampus of mice. Furthermore, the increased serum corticosterone level was also attenuated by the PFEO treatment. A total of 21 volatile compounds were detected in PFEO using GC-MS, among which elemicin (15.52%), apiol (15.16%), and perillaldehyde (12.79%) were the most abundant ones. The PFEO compounds targeted 32 depression-associated genes, which were mainly related to neural cells and neurotransmission pathways. Molecular docking indicated good binding affinities between the bioactive components of PFEO (apiol, β-caryophyllene, elemicin, and myristicin) and the key targets, including ACHE, IL1B, IL6, MAOB, SLC6A2, SLC6A3, SLC6A4, and tumor necrosis factor. Among the four compounds, β-caryophyllene, elemicin, and myristicin were more effective in reducing neurotoxicity and neuroinflammation. Elemicin showed the highest BBB permeability rate.

Conclusions: This study shows the antidepressant activities of PFEO in an SDS-induced mouse model and suggests its potential mechanisms of action: regulation of the corticosterone levels, hippocampal neurotransmitters, and ERK signaling. Apiol, β-caryophyllene, elemicin, and myristicin may be the main contributors to the observed effects induced by PFEO. Further studies are needed to fully elucidate the underlying mechanisms and the main PFEO bioactive components.

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List of abbreviations

5-HT	Serotonin
ACHE	Acetylcholinesterase
BBB	Blood-brain barrier
CORT	Corticosterone
DA	Dopamine
EPM	Elevated plus maze
FST	Forced swimming test
GABA	Gamma-aminobutyric acid
GC-MS	Gas chromatography-mass spectrometry
GO	Gene Ontology
HPA	Hypothalamic-pituitary-adrenal
IL	Interleukin
KEGG	Kyoto Encyclopedia of Genes and Genomes
MAOB	Monoamine oxidase B
NE	Norepinephrine
OFT	Open field test
PFEO	Perillae Folium essential oil
SDS	Social defeat stress
SLC6A2	Solute carrier family 6 member 2
TNF	Tumor necrosis factor
TST	Tail suspension test
TCMSP	Traditional Chinese Medicine Systems Pharmacology Database and Analysis Platform

1. Introduction

Depression is a common mental illness which is defined by low mood, lack of interest and pleasure in most activities, and suicidal ideation in severe cases (Kennedy, 2008). Owing to the COVID-19 outbreak, the incidence of depression in the general population has increased from 3.4% in 2017 to 25% in 2020 (Bueno-Notivol et al., 2021). Depression shows high comorbidity with other psychological diseases, particularly anxiety and substance use disorder, which increases its severity and the risk of recurrence, and decreases the quality of life of patients (Steffen et al., 2020). Moreover, comorbid depression is prevalent in patients with chronic diseases, including cardiovascular diseases, neurological disorders, diabetes, and cancer (Gold et al., 2020). Comorbid depression is linked to poorer prognosis and health scores and increased fatality risk in patients with somatic disorders (Koyanagi et al., 2018; Moussavi et al., 2007). Selective serotonin reuptake inhibitors (SSRIs) are the most commonly prescribed antidepressants (Wang et al., 2018). However, the long-term use of SSRIs, such as fluoxetine, citalopram, and paroxetine, is associated with various adverse events, including upper gastrointestinal bleeding, nausea, insomnia, nervousness, and cardiovascular side effects (Brambilla et al., 2005; de Abajo et al., 1999; Wang and Pae, 2013). Therefore, the development of alternative and complementary antidepressants with high efficacy and fewer adverse effects is necessary.

Several animal models of depression, such as chronic unpredictable mild stress (CUMS) and social defeat stress (SDS), have been established to identify antidepressants (Duman, 2010; Krishnan and Nestler, 2011). The CUMS model comprises a model that is repeatedly exposed to different unpredictable stressors, such as wet bedding, food deprivation, or restraint stress (Xie et al., 2015). The SDS model is a model repeatedly exposed to intruders or unfamiliar residents to induce a social subordination response (Golden et al., 2011). These models exhibit depression-like behaviors and have been widely used in depression studies. However, because social stress is considered the most common stressor in humans, the SDS-induced mouse model might be more relevant to study clinical depression and may be more advantageous than other animal models for investigating and developing antidepressant drugs (Carnevali et al., 2020). SDS induces social avoidance, anhedonia, and anxiety behaviors; however, the treatment of SDS models with antidepressants, such as venlafaxine and tianeptine, improves these behavioral impairments, suggesting the validity of the SDS model (Venzala et al., 2012).

Herbal medicine has been commonly used as complementary and alternative medicine for treating clinical depression with few adverse events (Haller et al., 2019; Nahas and Sheikh, 2011). Perillae Folium, the leaves and twigs of *Perilla frutescens* (L.) Britton, a commonly used vegetable and medicinal plant in Asia, is used in several traditional Chinese medicine herbal prescriptions, such as the Banxia Houpu Decoction and Xiang-su-san, for treating depression in China (Feng et al., 2016; Li et al., 2021; Luo et al., 2000). Fermented *P. frutescens* is known to alleviate depressive behavior in sleep-deprived mice by modulating the hypothalamic-pituitary-adrenal (HPA) axis and brain-derived nerve growth factor signaling pathways (Jee et al., 2023). However, the effects of the essential oil from Perillae Folium (PFEO) on an SDS-induced depression model and the mechanisms underlying its antidepressant effects have not been investigated. Therefore, this study explored the antidepressant mechanisms of PFEO using an SDS-induced depression mouse model and network pharmacology analysis.

2. Materials and methods

2.1. Preparation of the essential oil from Perillae Folium

Perillae Folium was purchased from Omniherb (Daegu, Republic of Korea) and validated by Professor In-Jun Yang (Dongguk University, Gyeongju, Republic of Korea) (voucher specimen: 2022-A-37). The herb was immersed in distilled water (1:10, w/v) in a 1,000 mL flask. The simultaneous distillation extraction (SDE) method was used to obtain PFEO using n-hexane as the extraction solvent. The SDE was carried out for 4 h, and n-hexane was removed using a vacuum evaporator. The final essential oil (yield: 0.39%, w/w) was kept at −20 °C for the subsequent experiments.

2.2. Animals

C57BL/6 mice (male, five-week-old) and male Institute of Cancer Research (ICR) retired breeder mice were purchased from Koatech (Gyeonggi, Republic of Korea) and acclimatized for a week before performing the experiments. All animal procedures were approved by the Institutional Animal Care and Use Committee (IACUC) of Dongguk University (Gyeongju, Republic of Korea) (approval no. IACUC-2023-18). The mice were maintained under normal conditions at 22 °C ± 2 °C, at a humidity of 50% ± 5%, and under a 12 h light/dark cycle.

2.3. Toxicity evaluation

The mice were divided into three groups ($n = 5$/group): control mice (the CON group), mice treated with 12.5 mg/kg of PFEO (the PFEO12.5 group), and mice treated with 25 mg/kg of PFEO (the PFEO25 group). The sample size aligns with the "resource equation" method (E = 12), falling within the recommended range (E = 10–20) and being supported by previous murine toxicity studies (Armah et al., 2021; Charan and Kantharia, 2013; Dantas-Medeiros et al., 2021). The PFEO doses were selected based on previous studies and our screening study that investigated the antidepressant effects of essential oils at doses of 12.5–50 mg/kg in mice (Rahmouni et al., 2015; Tran et al., 2023). PFEO was dissolved in 3% Tween 80 in saline and administered intranasally (10 μL/mouse) daily for three weeks. The body weight and weight of the organs (heart, lungs, kidneys, liver, and spleen) of each mouse were recorded. The nasal mucosa and lungs were collected, fixed in 4% paraformaldehyde, and then embedded in paraffin. Sections of the paraffin-embedded tissues were cut and stained with hematoxylin and eosin (H&E) for histological evaluation. Serum was collected by centrifuging the blood samples at 3,000 rpm for 15 min. The serum levels of glutamic pyruvic transaminase (GPT) and glutamic oxaloacetic transaminase (GOT) were measured using Fuji Dri-Chem Slide (Fujifilm, Tokyo, Japan) according to the manufacturer's instructions.

2.4. Social defeat stress (SDS) model and treatments

The mice were divided into five groups ($n = 8$/group): the CON group, SDS-exposed mice (the SDS group), SDS mice treated with 12.5 mg/kg of PFEO (the PFEO12.5 group), SDS mice treated with 25 mg/kg of PFEO (the PFEO25 group), and SDS mice treated with 3 mg/kg of memantine (the MEM group). PFEO was dissolved in 3% Tween 80 in saline and administered intranasally (10 μL/mouse). Memantine (Sigma-Aldrich, St. Louis, MO, USA) was dissolved in saline and intraperitoneally administered (3 mg/kg) as previously described (Almeida et al., 2006). Mice were treated with the drugs 30 min before SDS induction or behavioral testing. The experimental schedule of this animal study is presented in Supplementary Fig. S1.

Before the SDS procedure, male ICR mice (eight-month-old) were screened for aggressive behavior as previously described, with minor modifications (Feng et al., 2020). For screening, C57BL/6 mice were placed in the home cages of ICR mice for 3 min/day for three days. The selection criteria included ICR mice attacking C57BL/6 mice for two consecutive days with a latency to attack <1 min. During the SDS procedure, C57BL/6J mice were placed in the home cage of the ICR aggressor for 5 min for physical contact, followed by 24 h of visual contact by placing the cage of the C57BL/6J mice next to the cage of the ICR mice, as previously described (Kamimura et al., 2021). In the CON group, physical and visual contacts were induced using C57BL/6J mice instead of ICR mice. All the mice were single-housed during the experiments. Behavioral tests were conducted after 10 days of SDS induction. On the day after the last behavioral test, all mice were euthanized via isoflurane inhalation. Blood and brain samples were collected. Both hippocampi of each mouse were dissected for further analysis. The blood samples were centrifuged at 3,000 rpm for 15 min for serum collection.

2.5. Social interaction test (SIT)

The SIT was used to evaluate the effects of PFEO on the social behavior of SDS-exposed mice, as previously described (Golden et al., 2011). The SIT was performed in a plastic box (40 cm × 40 cm × 40 cm) containing a wire cage (10 cm × 10 cm × 15 cm) in the middle of one side of the box (Fig. 2A). The SIT consisted of two phases (150 s/phase; interval, 1 min), with the first one not including ICR mice in the wired cage (no target), and the second one including ICR mice in the wired cage (with target). The time spent in the social zone (8-cm wide areas around the wire cage) in both phases was assessed using the SMART video tracking software v.3.0 (Panlab, Barcelona, Spain). The SI ratio was determined by dividing the time spent in the social zone in the "with target" phase by the time spent in the social zone in the "no target" phase. Mice were categorized as resilient (SI ratio ≥1) or susceptible (SI ratio <1).

2.6. Open field test (OFT)

The OFT was used to assess the effect of PFEO on the locomotor activity and anxiety-like behavior of SDS-exposed mice, as previously described (Berton et al., 2021). Mice were placed in the middle of an OFT box (30 cm × 30 cm × 30 cm) and free exploration was allowed for 5 min. The time in the center (s), distance traveled in the center (cm), and total distance (cm) in the OFT were measured using the SMART software.

2.7. Elevated plus maze (EPM) test

The EPM test was used to assess the effects of PFEO on the anxiety-like behavior of SDS-exposed mice, as previously described (Walf and Frye, 2007). Mice were placed in the central zone of the EPM that comprised two open arms (5 cm × 30 cm) and two closed arms (5 cm × 30 cm × 15 cm) and free exploration was allowed for 5 min. The time and distance traveled in the open arms were measured using the SMART video tracking system.

2.8. Tail suspension test (TST)

The TST was conducted to assess the effects of PFEO on the depression-like behavior of SDS-exposed mice, as previously described (Can et al., 2012b). An adhesive tape (length, 15 cm) was used to attach the tail of each mouse to a bar 50 cm above the floor. The SMART software was used to track the immobility time of each mouse for 6 min.

2.9. Forced swimming test (FST)

The FST was performed to assess the antidepressant effects of PFEO in SDS-exposed mice, as previously described (Can et al., 2012a). Each mouse was placed in a plastic cylinder with a diameter of 20 cm and a height of 30 cm filled up with water at 25 °C to a level of 10 cm. The SMART software was used to track the immobility time of each mouse for 6 min.

2.10. Western blot analysis

Each hippocampus was homogenized in a lysis buffer (Tissue Extraction Reagent I; Thermo Fisher Scientific, Waltham, MA, USA). Proteins were separated using sodium dodecyl sulfate-polyacrylamide gel electrophoresis and transferred to polyvinylidene fluoride membranes (Merck Millipore, Carrigtwohill, Ireland). The membranes were blocked using 5% skim milk and incubated with the primary and secondary antibodies. The primary antibodies used included rabbit anti-phosphorylated extracellular signal-regulated kinase (p-ERK) (Cell Signaling Technology, Danvers, MA, USA; #4370, 1:2,500), rabbit anti-ERK (Cell Signaling Technology, #9102, 1:2,500), and mouse anti-β-actin (Sigma-Aldrich, A1978, 1:10,000). The secondary antibodies used included horseradish peroxidase-conjugated anti-rabbit IgG (Cytiva, NA934-1 ML, 1:5,000) or goat anti-mouse IgG (Enzo, BML-SA204-0100, 1:10,000) antibody. The protein bands were developed using enhanced chemiluminescence reagents and visualized using a ChemiDoc imager (Bio-Rad Laboratories, Hercules, CA, USA), and the intensity of the bands was measured using GelPro v.3.1 software (Media Cybernetics, Rockville, MD, USA).

2.11. Gas chromatography-mass spectrometry (GC-MS) analysis

The PFEO composition was analyzed using a GC-MS system (8890GC/5977MSD, Agilent, Santa Clara, CA, USA) and the corresponding instrument parameters are listed in Supplementary Table S1. Samples at a concentration of 512.45 mg/mL were prepared for analysis. The chemical constituents of PFEO were determined using the National Institute of Standards and Technology Mass Spectra Library.

2.12. Screening for the antidepressant targets of the PFEO compounds

The Traditional Chinese Medicine Systems Pharmacology Database and Analysis Platform (TCMSP) (https://old.tcmsp-e.com/tcmsp.php/, accessed on 09 January 2023) was used to assess the targets of the volatile compounds of PFEO (Ru et al., 2014). The DisGeNET database (http://www.disgenet.org/, accessed on January 09, 2023) and

Therapeutic Target Database (TTD) (https://db.idrblab.net/ttd/, accessed on January 09, 2023) were used to identify the depression-associated targets (DisGeNET disease ID: C0011570; TTD disease ID: ICD-9:311) (Piñero et al., 2017; Zhou et al., 2022). The PFEO components and depression-associated targets that overlapped were selected for pharmacological network analysis.

2.13. Pharmacological network construction

A compound-target network was constructed and analyzed using Cytoscape v.3.8.2 (Cytoscape Consortium, San Diego, CA, USA) (Shannon et al., 2003). The compounds and targets are indicated by nodes, and their interactions are indicated by edges. The topological features of the compounds in the network were analyzed. The Search Tool for the Retrieval of Interacting Genes/Proteins (STRING) (https://string-db.org/, accessed on January 09, 2023) was used to construct a protein-protein interaction (PPI) network of the common targets of both PFEO compounds and depression-related biomarkers (Timalsina et al., 2014).

2.14. Enrichment analysis

The Enrichr tool (https://maayanlab.cloud/Enrichr/, accessed on January 09, 2023) was used to analyze the Gene Ontology (GO) functions and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways of the possible antidepressant targets of PFEO compounds (Kuleshov et al., 2016). GO functional enrichment analysis consisted of three categories, including the biological process (BP), cellular component (CC), and molecular function (MF). The top 10 GO terms and KEGG pathways with a *p*-value <0.01 were chosen, and plots were generated using the SRplot tool (http://www.bioinformatics.com.cn/srplot, accessed on January 09, 2023) (Janani and Priya, 2022).

2.15. Molecular docking analysis

The interaction between the PFEO components and target proteins was estimated using AutoDock Vina version 1.2.0 (The Scripps Research Institute, CA, USA) (Trott and Olson, 2010). For docking analysis, the three-dimensional (3D) structures of the target proteins were downloaded from the RCSB Protein Data Bank (RCSB PDB) (https://www.rcsb.org/, accessed on January 11, 2023) and AlphaFold Protein Structure Database (https://alphafold.ebi.ac.uk/, accessed on January 11, 2023) (Burley et al., 2019) and the 3D structures of the PFEO compounds were acquired from the PubChem database (https://pubchem.ncbi.nlm.nih.gov/, accessed on January 11, 2023). The docking scores (binding affinities) and binding modes between the PFEO compounds and target proteins were automatically identified using AutoDock Vina. The 2D diagrams of the compound-target interaction were visualized using the Discovery Studio Visualizer v21.1 program (Sharma et al., 2021).

2.16. Cell culture and treatments

PC12 cells were cultured in RPMI media supplemented with 10% FBS (Merck KGaA, Darmstadt, Germany) and 1% penicillin-streptomycin (Thermo Fisher Scientific) at 37 °C and 5% CO_2. The cells were treated with apiol (CFN70479), β-caryophyllene (CFN90502), elemicin (CFN90685), and myristicin (CFN90589) (0.1, 1, 10, or 50 μM; Chem Faces, Wuhan, China) with or without corticosterone (CORT, 200 μM; 27840, Sigma-Aldrich, Seoul, South Korea) for 24 h.

BV2 cells were incubated in DMEM media supplemented with 10% FBS (Merck KGaA) and 1% penicillin-streptomycin (Thermo Fisher Scientific) at 37 °C and 5% CO_2. The cells were treated with apiol, β-caryophyllene, elemicin, and myristicin (50 or 100 μM) and dexamethasone (10 μM; D4902, Sigma-Aldrich, USA) with or without lipopolysaccharide (LPS, 1 μg/mL; L2880, Sigma-Aldrich, South Korea) for 24 h.

2.17. Water-soluble tetrazolium salt (WST) and lactate dehydrogenase (LDH) assays

The effects of the treatments on the viability of PC12 cells and their LDH release were examined using EZ-Cytox and EZ-LDH assay kits (DoGenBio, Seoul, Korea) by following the manufacturer's instructions. The absorbance was measured at 450 nm with a reference wavelength of 650 nm using an iMark microplate reader (Bio-Rad).

2.18. Enzyme-linked immunosorbent assay (ELISA)

An hippocampus of each mouse was homogenized in a lysis buffer (Tissue Extraction Reagent I; Thermo Fisher Scientific). The levels of serotonin (5-HT), dopamine (DA), norepinephrine (NE), and gamma-aminobutyric acid (GABA) in the hippocampus were determined using ELISA kits (Biomatik, Ontario, Canada) according to the manufacturer's instructions. The serum level of CORT was measured using an ELISA kit (Wuhan Fine Biotech Co., Ltd., Hubei, China). The levels of tumor necrosis factor (TNF)-α and IL-6 in the BV2 cell supernatant were analyzed using ELISA kits (LABISKOMA, Seoul, South Korea) by following the manufacturer's protocols. The absorbance was measured at 450 nm using a microplate reader (Bio-Rad Laboratories).

2.19. Parallel artificial membrane permeability assay (PAMPA)

The ability of apiol, β-caryophyllene, elemicin, and myristicin to penetrate the blood-brain barrier (BBB) was measured using a PMBBB-096 kit (BioAssay Systems, Eching, Germany) by following the manufacturer's protocol. In brief, 200 μL of a donor solution (500 μM) was placed over a brain lipid-containing membrane and then incubated for 18 h at 25 °C. Promazine hydrochloride and diclofenac were used as the high and low controls, respectively. The compounds at 200 μM were used as equilibrium standards. The absorbance was measured using a Spectramax M2 spectrophotometer (Molecular Devices, LLC, CA, USA) at the peak absorbance of each test compound.

2.20. Statistical analysis

Data were analyzed using GraphPad Prism 9.0 (GraphPad Software, San Diego, CA, USA) and are presented as the mean ± standard error of the mean (SEM) of at least three independent experiments. The differences between two groups were analyzed using an unpaired Student's *t*-test or one-way analysis of variance (ANOVA) with Bonferroni's *post-hoc* test. For non-normally distributed data (those that did not pass the Anderson-Darling and Shapiro-Wilk normality tests), the Kruskal-Wallis test with uncorrected Dunn's test was used for comparison. A $p < 0.05$ was considered statistically significant.

3. Results

3.1. Toxicity evaluation

As shown in Fig. 1A, after three weeks of treatment, no significant differences in the body weight between the CON group and the PFEO-treated group mice were observed. Moreover, the intranasal administration of PFEO did not alter the organ indices, including those of the heart, lungs, spleen, liver, and kidneys (Fig. 1B). The serum levels of GPT and GOT were not significantly different among the groups (Fig. 1C). The nasal mucosa of mice in all groups exhibited a normal and intact epithelial layer with ciliated cells. No significant signs of tissue damage or inflammation are evident (Fig. 1D). In addition, the lung tissues of the treated group mice appear healthy, with an intact alveolus and bronchiole structural integrity (Fig. 1E).

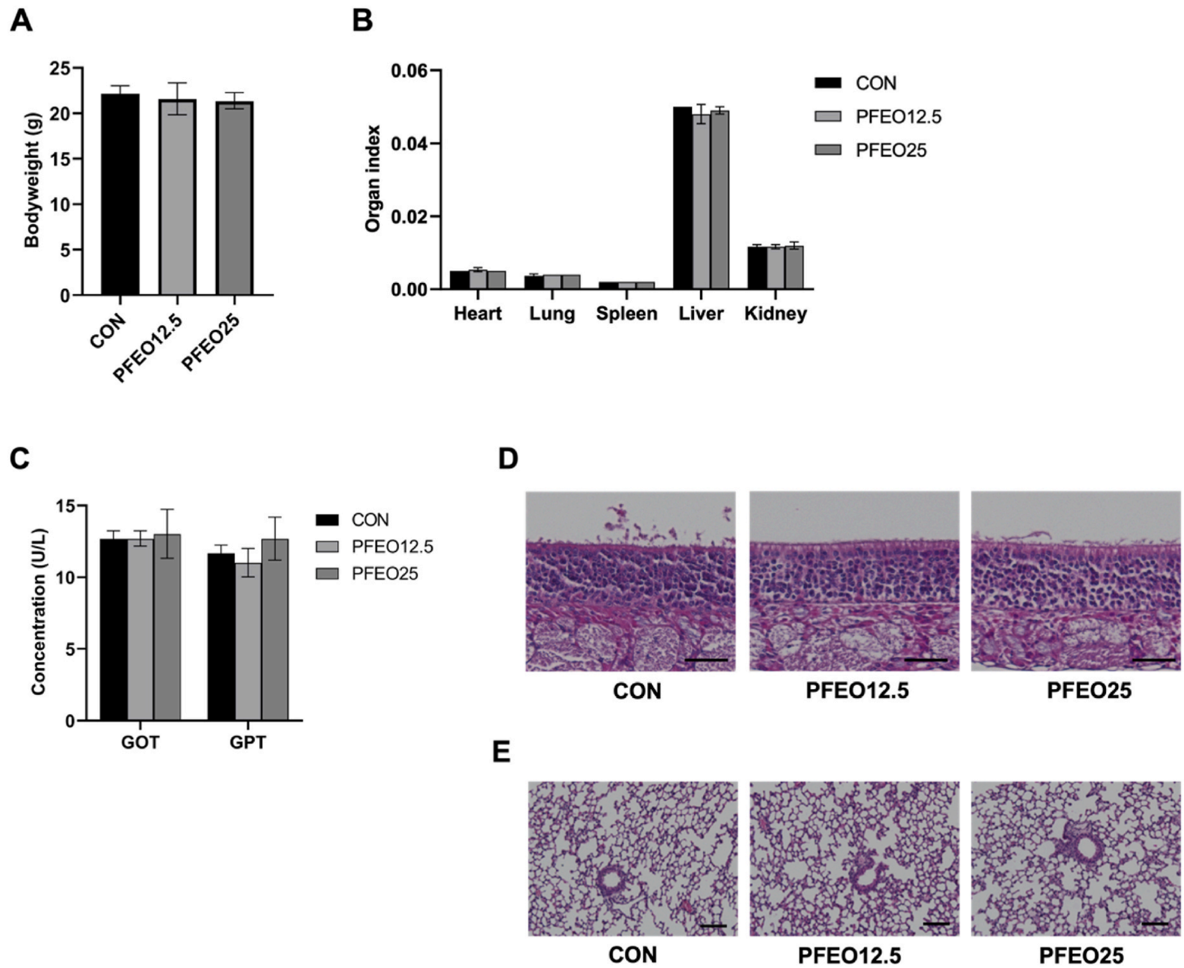

Fig. 1. Toxicity evaluation results. (A) Body weight of the mice. (B) Organ indices of the mice. (C) Serum GPT and GOT levels in the mice. (D) H&E staining results of the nasal mucosa (magnification: 20 ×; scale bar: 100 μm). (E) Results of the H&E staining of the lungs (magnification: 10 ×; scale bar: 100 μm). Data are presented as the mean ± SEM (n = 5 per group).

3.2. Effect of PFEO on the social behavior of SDS-exposed mice in the SI test

A SIT was used to evaluate the effects of PFEO on the social behavior of SDS-exposed mice (Fig. 2A). Fig. 2B and C indicate that the time in the social zone and the SI ratio in the SIT were significantly lower in the SDS group than in the PFEO treatment and control groups ($p < 0.05$). In contrast, the intranasal administration of PFEO (12.5 and 25 mg/kg) improved the SDS-induced decreases in the time in the social zone and the SI ratio ($p < 0.05$) (Fig. 2C). In addition, the percentages of susceptible and resilient mice were also improved by the PFEO treatment (Fig. 2D).

3.3. Effect of PFEO on the locomotor activity and anxiety-like behavior of SDS-exposed mice according to the OFT

An OFT was conducted to evaluate the effects of the intranasal administration of PFEO on the locomotor activity and anxiety-like behavior of SDS-induced mice. In the SDS group, the time and distance traveled in the center, as well as the total distance traveled, were significantly lower than those in the CON group ($p < 0.05$) (Fig. 3). In contrast, in the PFEO-treated groups (12.5 and 25 mg/kg), these parameters were higher than those in the SDS group ($p < 0.05$).

3.4. Effect of PFEO on the anxiety-like behavior of SDS-exposed mice according to the EPM test

An EPM test was performed to assess the effect of PFEO on the anxiety-like behavior of SDS-exposed mice. SDS induced significant decreases in the time spent and distance traveled in the open arms (Fig. 4). In contrast, the intranasal administration of PFEO (12.5 and 25 mg/kg) significantly attenuated the reduction in these parameters ($p < 0.05$).

3.5. Effect of PFEO on the depression-like behavior of SDS-exposed mice

A TST and a FST were performed to examine the effect of PFEO on the SDS-induced depression-like behavior of mice. Fig. 5A shows that the immobility time of the mice to which PFEO (12.5 and 25 mg/kg) was intranasally administered was significantly lower than that of the SDS-induced mice ($p < 0.05$), according to the TST results. Similarly, the immobility time of the PFEO12.5 group mice was lower than that in the SDS group mice ($p < 0.05$), according to the FST results (Fig. 5B).

3.6. Effects of PFEO on the levels of neurotransmitters in the hippocampus and of CORT in the serum of SDS-exposed mice

The effect of the intranasal administration of PFEO on the hippocampal levels of neurotransmitters and the serum levels of CORT in SDS-exposed mice was examined. As shown in Fig. 6A, SDS induced

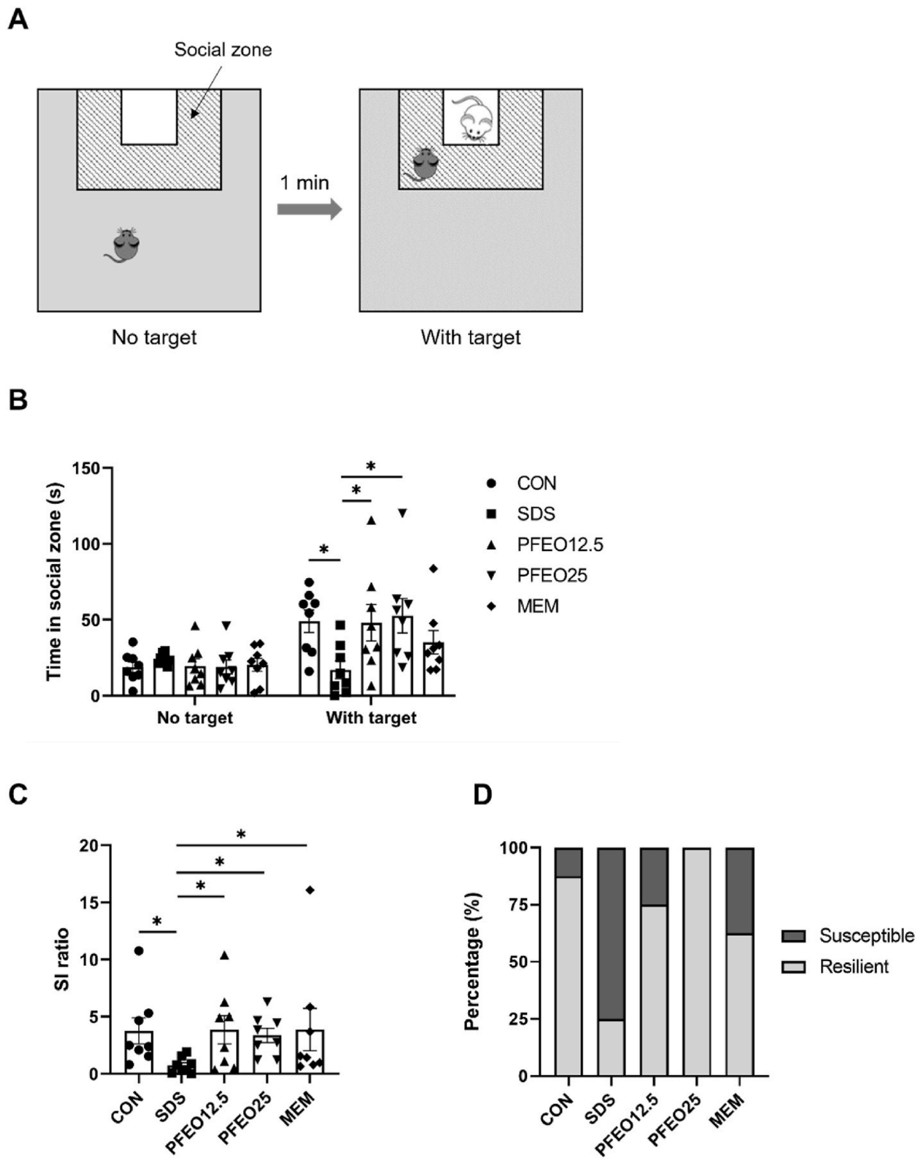

Fig. 2. Effect of PFEO on the social behavior of SDS-exposed mice according to the SI test. (A) SI test procedure. (B) Time spent in the social zone for each group. (C) SI ratio for each group. (D) Susceptible and resilient mice percentages in the five groups. Data are presented as the mean ± SEM ($n = 8$ per group). *$p < 0.05$.

significant decreases in the levels of 5-HT and DA in the hippocampus compared to those of the control group ($p < 0.05$). SDS exposure also reduced, but not significantly, the hippocampal GABA and NE levels compared to those of the control group. In contrast, PFEO (12.5 and 25 mg/kg) significantly improved the hippocampal levels of 5-HT, GABA, and NE ($p < 0.05$). However, PFEO did not affect the hippocampal DA level. As shown in Fig. 6B, the serum CORT levels were significantly higher in the SDS group than in the CON group ($p < 0.05$). In contrast, the intranasal administration of PFEO (12.5 mg/kg) significantly decreased the SDS-induced increase in the serum CORT levels ($p < 0.05$).

3.7. Effects of PFEO on the ERK signaling pathway in SDS-exposed mice

Previous studies have indicated that the ERK signaling pathway may contribute to an SDS-induced depression-like behavior in mice (Iio et al., 2011; Ji et al., 2022). Thus, here, the effect of the intranasal administration of PFEO on the ERK signaling pathway in SDS-exposed mice was examined. As shown in Fig. 7, SDS induced a significant decrease in the hippocampal p-ERK levels ($p < 0.05$). In contrast, PFEO (12.5 and 25 mg/kg) significantly increased the phosphorylation of ERK ($p < 0.05$).

3.8. Chemical composition of PFEO

The chemical constituents of PFEO were analyzed using GC-MS (Fig. 8 and Table 1). A total of 21 volatile compounds were detected in PFEO. Elemicin (15.52%), apiol (15.16%), perillaldehyde (12.79%), and β-caryophyllene (7.39%) were the most abundant PFEO components identified (Table 1).

3.9. Pharmacological network construction

Among the 21 volatile PFEO compounds identified, seven of them had no targets in the TCMSP database, and the others had 52 corresponding targets (Supplementary Table S2). A total of 1,498 depression-associated targets were identified using the DisGeNET and TTD databases. A total of 32 common targets were found between the PFEO and depression-associated targets (Supplementary Fig. S2A). Cytoscape v.3.8.2 was used to visualize the compound-common target network. The network contained 46 nodes (14 compounds and 32 targets) and

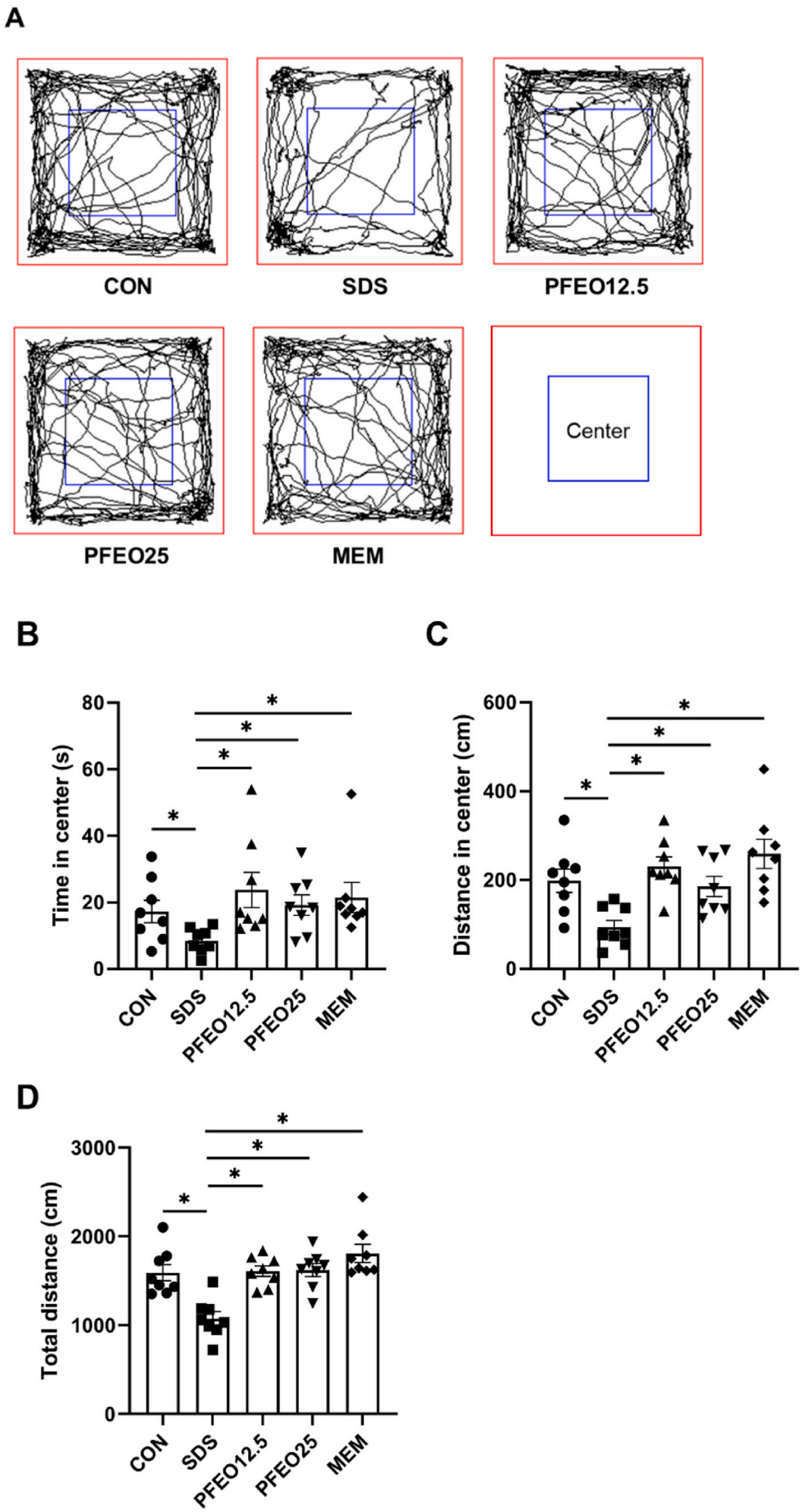

Fig. 3. Effect of PFEO on the locomotor activity and anxiety-like behavior of SDS-exposed mice in the OFT. (A) Representative traveling paths in the OFT. (B) Time spent in the center for each of the groups. (C) Distance traveled in the center for each of the groups. (D) Total distance traveled for each of the groups. Data are presented as the mean ± SEM ($n = 8$ per group). $^{*}p < 0.05$.

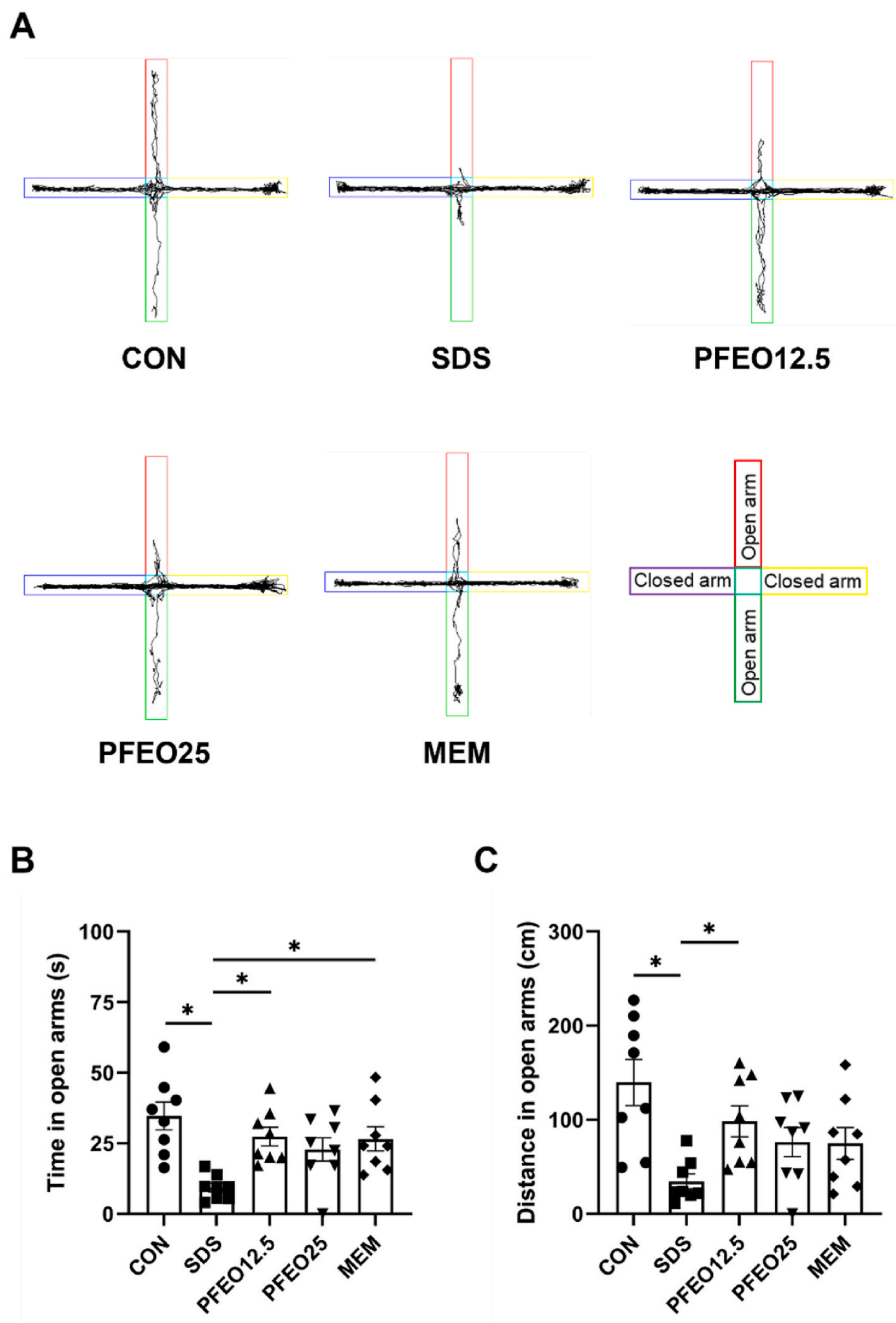

Fig. 4. Effect of PFEO on the anxiety-like behavior of SDS-exposed mice according to the EPM test. (A) Representative traveling paths observed in the EPM test. (B) Time spent in the open arms for each group. (C) Distance traveled in the open arms for each group. Data are presented as the mean ± SEM (n = 8 per group). *p < 0.05.

120 edges (Supplementary Fig. S2B). The topological analysis of the PFEO components is shown in Table 2. The key components with high degree, betweenness centrality, and closeness centrality, such as apiol, myristicin, elemicin, and β-caryophyllene, might be the main contributors to the antidepressant effect of PFEO. The PPI network of the common targets between the PFEO and depression-associated targets comprised 32 nodes and 113 edges (Supplementary Fig. S2C). The node degrees of solute carrier family 6 member 4 (SLC6A4), interleukin 6 (IL6), SLC6A3, and TNF were >10, implying that these targets might be crucial in mediating the anti-depression effects of PFEO (Supplementary Fig. S2D).

3.10. Enrichment analysis

Fig. 9A shows the GO enrichment results of the common targets between the PFEO and depression-associated targets, including the BP, MF, and CC. The main BP terms included the adenylate cyclase-activating adrenergic receptor signaling pathway, adrenergic receptor signaling pathway, and anterograde trans-synaptic signaling. The main MF terms were alpha-adrenergic receptor activity, epinephrine binding, and transmitter-gated ion channel activity involved in regulation of postsynaptic membrane potential. The most enriched CC terms included the integral component of the plasma membrane, neuron projection, and dendrite. The top 10 KEGG pathways are shown in Fig. 9B, and include

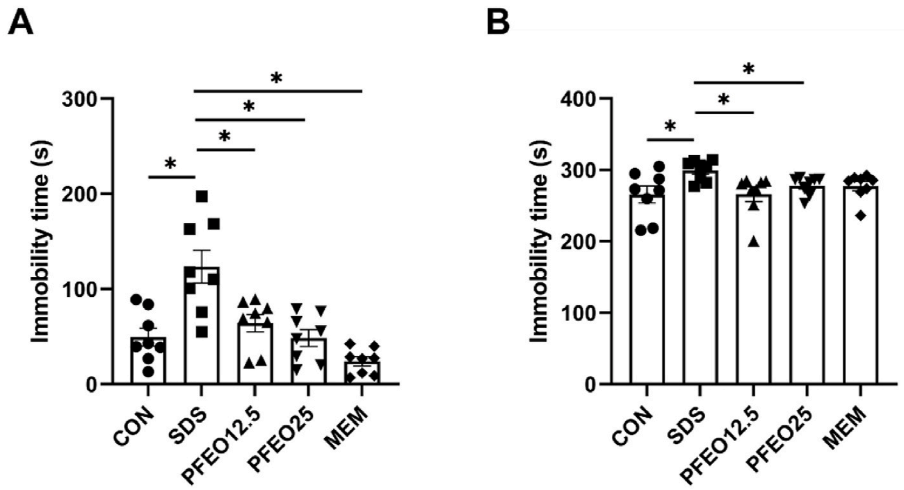

Fig. 5. Effect of PFEO on the depression-like behavior of SDS-exposed mice according to the TST and FST results. Immobility times obtained in the (A) TST and (B) FST. Data are presented as the mean ± SEM ($n = 8$ per group). $^*p < 0.05$.

the neuroactive ligand–receptor interaction, calcium signaling pathway, cyclic guanosine monophosphate (cGMP)–protein kinase G (PKG) signaling pathway, and serotonergic synapse. In addition, the regulation of the mitogen-activated protein kinase (MAPK) cascade and cyclic adenosine monophosphate (cAMP) signaling pathway might also contribute to the antidepressant effect of PFEO. The genes associated with the GO terms and KEGG pathways identified are listed in Supplementary Tables S3–S6.

3.11. Molecular docking analysis

Molecular docking analysis was performed to evaluate the binding affinities between the four active components of PFEO, apiol, myristicin, elemicin, and β-caryophyllene, and the potential target proteins (node degree ≥10 in the compound-target network and PPI network, respectively). Among the four PFEO active compounds, β-caryophyllene showed docking scores lower than −6.0 kcal/mol with almost all the potential protein targets, suggesting a strong binding affinity between them (Table 3). The interaction modes between the four compounds and target proteins are shown in Supplementary Fig. S3.

3.12. Effects of the PFEO compounds on CORT-induced neurotoxicity in PC12 cells

To test whether the PFEO compounds were toxic to PC12 cells, a WST assay was performed. All compounds at doses of 0.1, 1, 10, and 50 μM did not cause toxicity to PC12 cells (data not shown); these doses were chosen for the subsequent experiments. As shown in Fig. 10A, the CORT treatment at 200 μM for 24 h was toxic to PC12 cells, as evidenced by a cell viability reduction to 65% and a LDH release elevation to 270% in the CON group mice. The pretreatment with apiol, β-caryophyllene, elemicin, and myristicin dose-dependently increased the cell survival rate and reduced the LDH level ($p < 0.05$). These results suggest that apiol, β-caryophyllene, elemicin, and myristicin exerted a neuroprotective effect on the CORT-induced neurotoxicity to PC12 cells.

3.13. Effects of essential oils on LPS-induced neuroinflammation in BV2 cells

A WST assay was also performed to screen the non-toxic dose of the PFEO compounds in BV2 cells. The compounds at all doses (0.1, 1, 10, 50, and 100 μM) did not significantly decrease the cell viability (data not shown); therefore, the two highest doses, 50 and 100 μM, were used for the subsequent experiments. As shown in Fig. 10B, LPS (1 μg/mL) remarkably increased the TNF-α and IL-6 levels compared to those observed in the CON group mice; however, this overproduction was reduced by the β-caryophyllene and myristicin pretreatment in a dose-dependent manner ($p < 0.05$). The effect of β-caryophyllene was comparable with that of DEX (10 μM) ($p < 0.05$). Elemicin decreased only the IL-6 level, whereas no anti-inflammatory effect was observed for apiol.

3.14. BBB permeability of the PFEO compounds

The PAMPA assay was employed to assess the capacity of PFEO compounds to cross the BBB, and to compare it with that of the high control, promazine hydrochloride, and low control, diclofenac, provided in the PAMPA-BBB kit. Fig. 10C illustrates the permeability values, with apiol, β-caryophyllene, elemicin, and myristicin exhibiting values of 1.01×10^{-6}, 2.00×10^{-6}, 3.41×10^{-6}, and 0.97×10^{-6} cm/s, respectively. These findings indicate that apiol and myristicin have limited BBB permeability, whereas β-caryophyllene can moderately traverse the BBB, compared to diclofenac ($p < 0.05$). Notably, elemicin demonstrates a much higher BBB permeability than diclofenac ($p < 0.05$), being comparable to that of promazine hydrochloride.

4. Discussion

Essential oils and aromatherapy are commonly used alternative treatments for depression, with high efficacy and few side effects (Perry and Perry, 2006). A previous study has shown that the volatile oil from Xiang-Su, a herbal medicinal pair containing Perillae Folium and Cyperi Rhizoma, exerted antidepressant effects in menopausal rats (Li et al., 2021). Here, we demonstrated the antidepressant activity of the essential oil from Perillae Folium (PFEO), as well as its potential therapeutic mechanisms, using an SDS-induced depression mouse model. We also performed *in silico* analysis and validated a compound study to predict the most active components of PFEO.

The nasal-brain drug delivery pathway has attracted a lot of attention in drug development in recent years. While intranasal administration provides consistent absorption and allows for a more precise dosage calculation, the inhalation method cannot guarantee these attributes due to factors such as varying lung capacity, breathing patterns, and inhalation technique (Hedigan et al., 2023). Furthermore, both inhalation and oral administration could induce lung inflammation and cytotoxicity, even with acute treatment (Gosens et al., 2016; Rojas-Armas et al., 2019). Drugs administered intranasally can exert safe effects with high bioavailability by avoiding first-pass metabolism and by directly

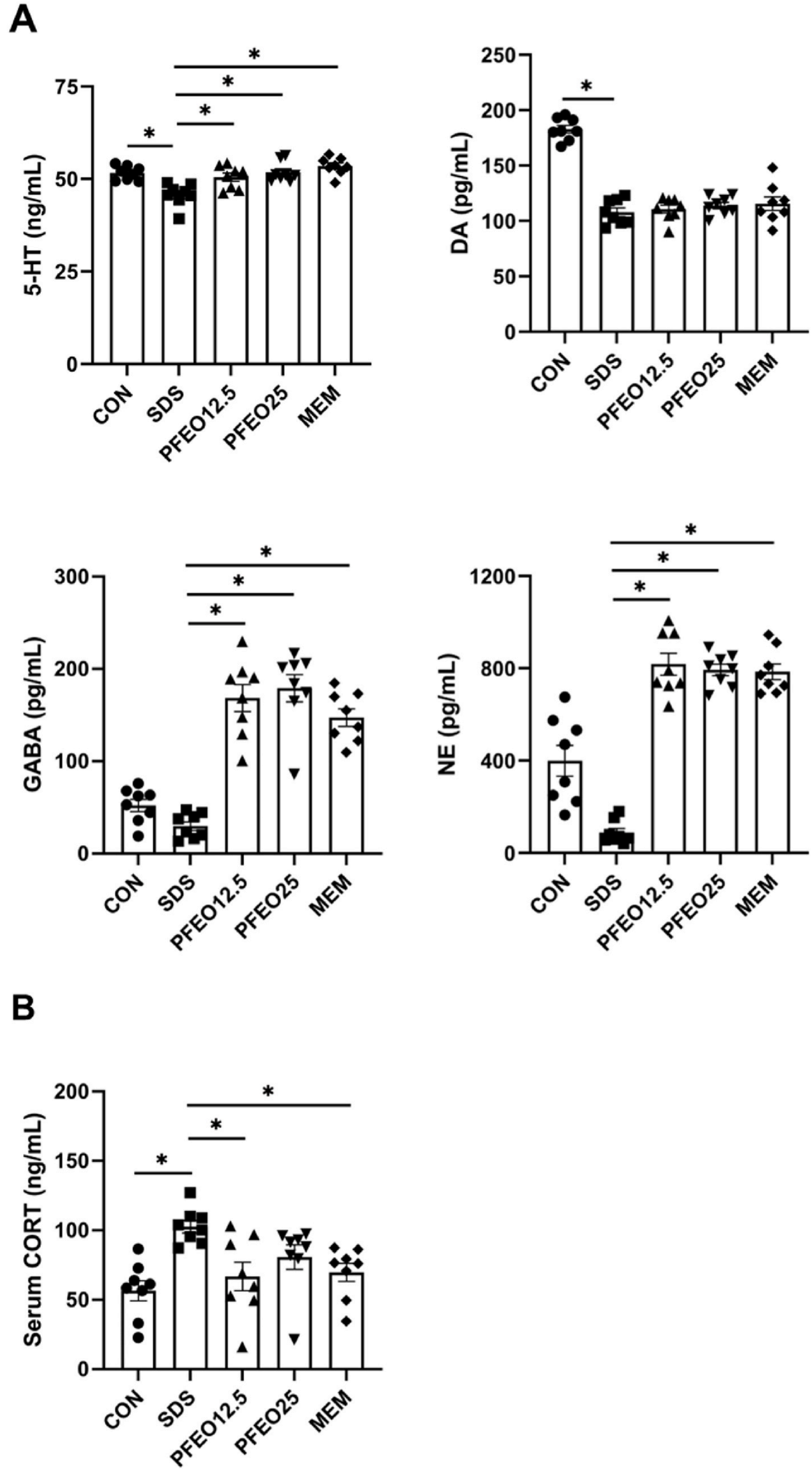

Fig. 6. Effects of PFEO on the levels of hippocampal neurotransmitters and on the serum CORT levels in SDS-exposed mice. (A) Hippocampal 5-HT, DA, GABA, and NE levels in the various groups. (B) Serum CORT level in the various groups. Data are presented as the mean ± SEM ($n = 8$ per group). $*p < 0.05$.

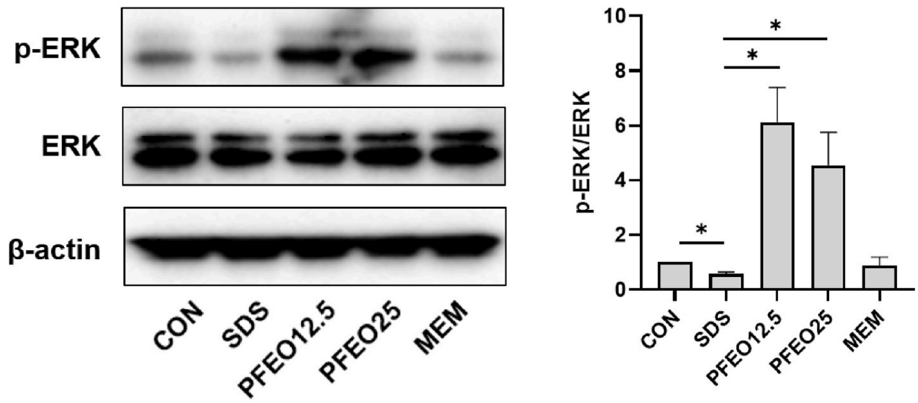

Fig. 7. Effects of PFEO on the ERK signaling pathway in SDS-exposed mice. The protein levels of p-ERK and ERK in the hippocampus were evaluated using western blot. Data are presented as the mean ± SEM ($n = 3$ per group). $*p < 0.05$.

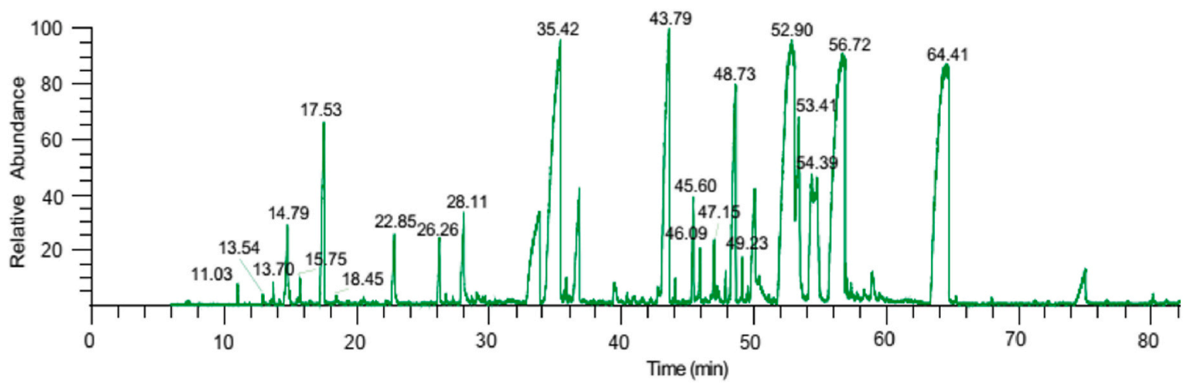

Fig. 8. GC-MS chromatogram obtained for PFEO.

Table 1
Chemical compounds of PFEO.

Compound	Retention time (min)	Relative content (%)
α-Thujene	11.03	0.11
Sabinene	13.54	0.03
β-Pinene	13.7	0.12
3-Octenol	14.79	0.92
3-Octanol	15.75	0.14
Limonene	17.53	2.41
Benzeneacetaldehyde	18.45	0.09
Linalool	22.85	0.76
Isomenthone	26.26	0.6
(±)-Menthol	28.11	1.38
Perillaldehyde	35.42	12.79
β-Caryophyllene	43.79	7.39
α-Humulene	45.6	0.89
(E)-β-Farnesene	46.09	0.41
β-Copaene	47.15	0.75
(Z,E)-α-Farnesene	48.73	3.89
Myristicin	49.23	0.31
Elemicin	52.9	15.52
Caryophyllene oxide	53.41	3.7
Allyltetramethoxybenzene	54.39	5.35
Apiol	56.72	15.16

crossing the BBB (Shringarpure et al., 2021). Here, we demonstrated that the intranasal administration of PFEO for 21 days did not show any toxic effects on the nasal mucosa and lungs of the treated mice. Therefore, we administered PFEO intranasally to SDS-induced mice to examine its antidepressant properties.

SDS is one of the best depression animal models for investigating and developing antidepressant drugs owing to its resemblance to social

Table 2
Topological features of the compounds in the compound-target network.

Compound	Degree	Betweenness Centrality	Closeness Centrality
Apiol	20	0.183147	0.535714
Myristicin	20	0.197281	0.535714
Elemicin	18	0.174222	0.511364
β-Caryophyllene	11	0.092008	0.441176
Caryophyllene oxide	7	0.095423	0.409091
α-Humulene	7	0.143664	0.416667
β-Pinene	7	0.021547	0.409091
Limonene	6	0.05742	0.409091
(E)-β-Farnesene	5	0.007546	0.394737
Perillaldehyde	5	0.028391	0.394737
Linalool	5	0.011623	0.394737
Menthol	4	0.023392	0.387931
Isomenthone	4	0.021912	0.387931
Benzeneacetaldehyde	1	0	0.241935

stress, which triggers depressive symptoms in humans (Carnevali et al., 2020). In accordance with previous studies, our results showed that SDS induced behavioral impairments, including increased social avoidance, as observed in the SIT, decreased motivation of escaping unpleasant situations, as observed in the TST and FST, or reduced exploration into brightly open and elevated spaces, as observed in the OFT and EPM (Iñiguez et al., 2016; Venzala et al., 2012). Anxiety is a common psychological comorbidity associated with depression, and over 80% of patients with depression have anxiety symptoms (Steffen et al., 2020; Tiller, 2013). Hence, assessing the effects of antidepressants through the anxious behavior observed in the OFT and EPM is also plausible (Montazeri et al., 2023). The intranasal administration of PFEO significantly improved the social interaction, as well as the depression- and

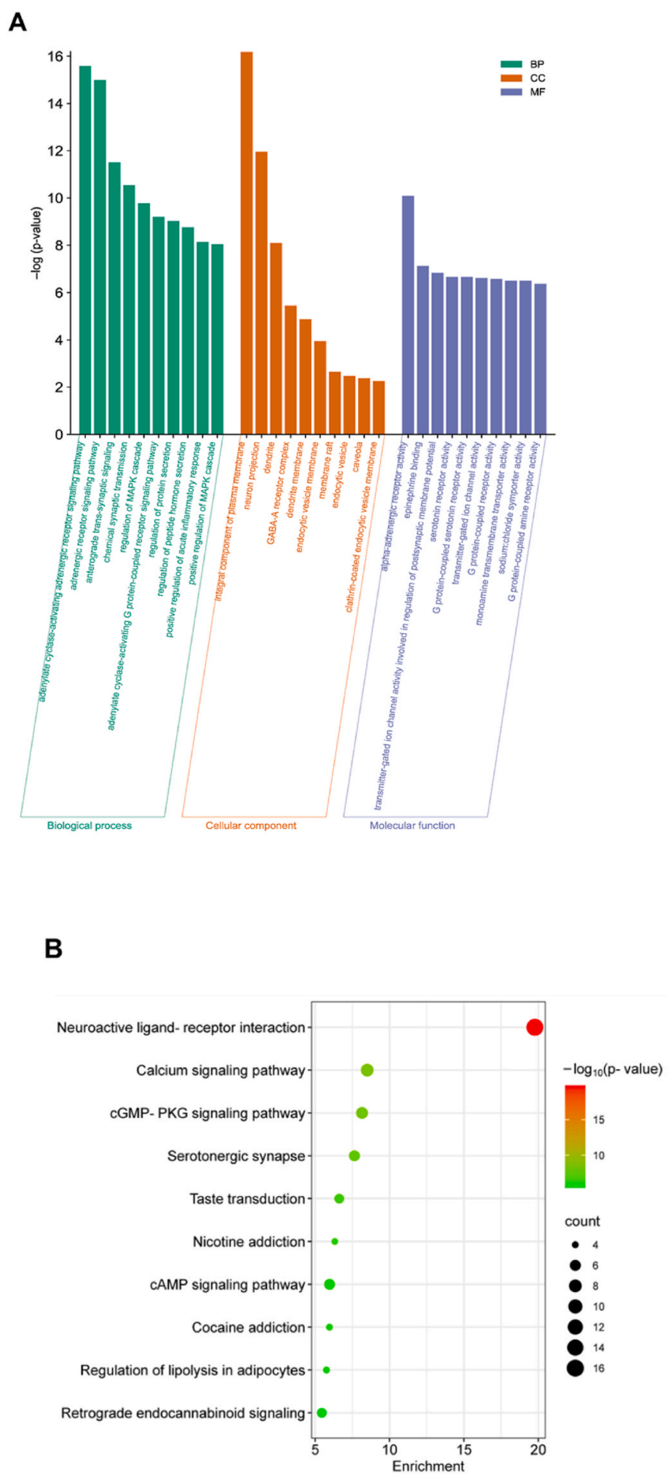

Fig. 9. GO and KEGG analyses of the common targets between those of PFEO compounds and the depression-associated ones. (A) GO enrichment analysis results. (B) KEGG pathway analysis results.

anxiety-like behaviors of SDS-induced mice, suggesting that PFEO exerted antidepressant effects against social stress.

Neurotransmitter and neuromodulator systems play important roles in the pathogenesis of psychiatric disorders, including depression (Bao et al., 2012; Inserra et al., 2021). Monoamine neurotransmitters, such as 5-HT, DA, and NE, function as chemical messengers in the brain that regulate mood and behavior (Nutt, 2008). Serotonin stabilizes mood and emotions, whereas dopamine governs pleasure and reward, and norepinephrine contributes to alertness and attention; hence, the disruption of the balance of these monoamine neurotransmitters could result in depression symptoms (Hannon and Hoyer, 2008; Herat et al., 2019; Lewis et al., 2021). Common medications for depression, including monoamine oxidase inhibitors, SSRIs, and serotonin-norepinephrine reuptake inhibitors, increase the levels of these monoamine neurotransmitters to ameliorate the depression symptoms (Ramachandraih et al., 2011). Amino acid neurotransmitters, such as GABA, also play a role in the development of depression. Decreased GABA levels and function, characterized by reduced inhibitory GABAergic activity and heightened excitatory neurotransmission, have been observed in patients with depression (Fee et al., 2017; Levinson et al., 2010). Moreover, the hyperactivation of the HPA axis with high levels of glucocorticoids may alter the neurotransmitter system (Stokes, 1995). The chronic exposure to corticosteroids is associated with lower levels of neurotransmitters, including 5-HT, DA, NE, epinephrine, glutamate, and GABA, and depression-like behavior (Huang et al., 2014). Hence, the bidirectional influence between neurotransmission and CORT regulation is closely involved in the physiopathology of depression. Intranasal PFEO administration reversed the SDS-induced depression behaviors, elevated serum CORT levels, and reduced the hippocampal levels of 5-HT, DA, NE, and GABA, suggesting that the antidepressant effects of PFEO might be mediated by the modulation of the stress hormone response and neurotransmitter systems. This also aligns with the neurotransmission pathways derived from the *in silico* predictions, such as epinephrine binding, serotonin receptor, adrenergic receptor, or GABA-A receptor signaling pathways. Moreover, the *in silico* identification of the cAMP signaling pathway, known for its regulatory role in corticosterone production, also provides a mechanistic link (Kang et al., 2023). The consistency between our in vivo and *in silico* experiment results supports the hypothesis that the action of PFEO may involve the modulation of neurotransmitters and the stress hormone cascade.

Interestingly, the PFEO treatment did not improve the DA levels, consistent with previous studies reporting that certain doses of essential oils exert beneficial effects on the production of 5-HT, NE, and GABA, but not on that of DA (Chen et al., 2023; Hao et al., 2013). Each neurotransmitter plays distinct roles and is subject to varying levels of regulation and feedback loops (Rosikon et al., 2023; Teleanu et al., 2022). Abnormalities in the dopamine system can contribute to specific features of depression, such as anhedonia; however, dopamine has been traditionally associated with other mental health conditions, such as schizophrenia and bipolar disorder, rather than depression (Juárez Olguín et al., 2016). PFEO is likely to selectively modulate the pathways associated with 5-HT, NE, and GABA production, but to not affect the dopaminergic pathways. Moreover, factors such as the brain regions targeted, or compositions, dosage, and duration of the PFEO treatment may all contribute to the differences in the neurotransmitter responses observed (Kumar, 2011; Tan et al., 2012).

ERK, a subclass of MAPKs, is involved in many biological processes, including neural plasticity, proliferation, and differentiation, in the central nervous system (Mazzucchelli & Brambilla, 2000; Roux & Blenis, 2004). The ERK/MAPK pathway plays important roles in the pathogenesis and treatment of depression (Gourley et al., 2008; Guan et al., 2013; Wang & Mao, 2019). The inhibition of ERK by PD184161 (an inhibitor of MAPK kinase) results in depression-like behavior in mice (Duman et al., 2007). ERK mediates the effects of antidepressants, such as fluoxetine and rapastinel, in chronic and unpredictable mild stress-induced mice (First et al., 2011; Shen et al., 2022). Previous studies have shown that the inhibition of the ERK/MAPK signaling pathway is associated with depression-like behavior in SDS-induced mice (Iio et al., 2011; Ji et al., 2022; Jiang et al., 2020). By connecting the observed improvement in ERK phosphorylation in SDS mice with the involvement of the MAPK signaling pathway predicted using GO enrichment analysis, our findings suggest that the potential antidepressant effects of PFEO may be mediated by partly modulating the

Table 3
Docking scores between the PFEO compounds and key target proteins.

Compounds	Docking score (kcal/mol)							
	ACHE	IL1B	IL6	MAO	SLC6A2	SLC6A	SLC6A	TNF
Apiol	−7.1	−5.1	−5.2	−5.5	−5.5	−5.4	−5.5	−6.0
β-Caryophyllene	−7.9	−5.6	−6.3	−6.2	−6.6	−6.7	−6.5	−7.0
Elemicin	−6.9	−4.9	−4.7	−5.1	−5.2	−5.6	−5.4	−5.8
Myristicin	−7.1	−5.1	−5.1	−5.7	−6.0	−6.4	−6.2	−6.0

ERK/MAPK signaling pathway.

GC-MS analysis revealed that PFEO contains 21 volatile compounds, with elemicin, apiol, perillaldehyde, and β-caryophyllene being the most abundant ones. These compounds are predicted to target various depression-related neurotransmitter receptors, consistent with the results obtained from animal experiments, which suggest that PFEO may exhibit antidepressant effects by enhancing the activity of 5-HT, GABA, and NE. Furthermore, the four compounds exhibited strong interactions with AChE, which is an enzyme responsible for regulating the acetylcholine neurotransmitter and is associated with depression-like behavior (Wang et al., 2019). These results imply that PFEO exerts its antidepressant effects by modulating neurotransmission pathways, with β-caryophyllene, elemicin, myristicin, and apiol being the main contributors to these effects.

The chronic exposure to elevated levels of corticosterone or neuroinflammation, characterized by microglial activation, is implicated in the pathogenesis of depression (Meng et al., 2019; Rhie et al., 2020). Therefore, CORT-treated PC12 cells and LPS-treated BV2 cells are commonly used to evaluate the potential antidepressant effects of candidate drugs (Lew et al., 2020; Park et al., 2020). The ability of a substance to traverse the BBB is also a pivotal determinant of its efficacy in treating neurological disorders, including depression (Wu et al., 2023). Apiol appears to lack both significant BBB permeability and anti-neuroinflammatory effects, suggesting that its contribution to the overall pharmacological impact of PFEO is limited. Interestingly, β-caryophyllene and myristicin exhibited optimal *in vitro* efficacy; however, their BBB permeability remained average. Conversely, elemicin, which has a high permeability index, did not significantly elevate the TNF-α levels in BV2 cells. The results further support the predictions obtained from the molecular docking analyses, indicating weak affinities between apiol or elemicin and the receptors associated with neuroinflammation, including IL1B, IL6, and TNF, as well as stronger binding scores between β-caryophyllene or myristicin and these receptors. Additionally, because nasal administration allows drugs to reach the brain through diverse pathways, these data led us to hypothesize that during intranasal PFEO treatment, β-caryophyllene and myristicin might be directly transported to the brain via the olfactory pathway, exerting their antidepressant effects by reducing inflammation and neurotoxicity (Tashima, 2020). In contrast, elemicin may follow a chemical penetration route, efficiently permeating the BBB through systemic circulation (Trevino et al., 2020). While empirical evidence for this hypothesis is lacking, our study contributes to understanding the pharmacokinetics of these compounds. The observed permeability profiles prompt us to consider the feasibility and necessity of the intranasal delivery of β-caryophyllene and myristicin, aiming to attain the ideal therapeutic concentrations of these compounds in the brain.

Our study used memantine as the positive control instead of an SSRI or tricyclic antidepressant for some reasons. First, memantine has been consistently reported to be well-tolerated, with minimal and mild side effects (Czarnecka et al., 2021; Krzystanek et al., 2021), whereas, as previously highlighted, the SSRI fluoxetine showed paradoxical effects in C57BL6/J mice, and existing antidepressants, such as fluoxetine and imipramine, exhibited no impact on the stress-induced corticosterone levels (David et al., 2009; Gosselin et al., 2017). Second, recent preclinical and clinical studies consistently support the efficacy of memantine in treating depression, since it is an N-methyl-D-aspartate receptor antagonist and can regulate glutamate neurotransmission, which is strongly implicated in the pathophysiology of depression (Hsu et al., 2022; Sani et al., 2012). Last and most importantly, memantine was reported to enhance stress resistance in a social defeat stress model (García-Pardo et al., 2019; Yoshida et al., 2022). Collectively, these results supported our selection of memantine as a positive control for this study. As expected, here, memantine treatment showed positive effects on SDS-exposed mice, as well as on depression-associated biomarkers.

Our research is still limited regarding the validation of the effects of the four compounds on depression models, as well as their influence on the activity of the recommended neurotransmitters based on the *in silico* analysis. Moreover, we only focused on the expression of neurotransmitters in the hippocampus, whereas other brain regions also play roles in the pathophysiology of depression and should be investigated. Therefore, further studies are required to better understand the mechanisms underlying the antidepressant effects of PFEO and its major bioactive components.

5. Conclusion

Our study showed that the intranasal administration of PFEO can improve social behavior and alleviate depression- and anxiety-like behaviors by regulating the levels of CORT, as well as those of hippocampal neurotransmitters and the ERK signaling pathway. Apiol, β-caryophyllene, elemicin, and myristicin might be the key components responsible for the antidepressant activity of PFEO by modulating neurotransmitter and neuroreceptor signaling.

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Authors' contributions

Conceptualization, I.-J.Y and L.T.H.N; investigation, L.T.H.N; formal analysis, L.T.H.N; writing- original draft preparation, L.T.H.N; writing-reviewing and editing, I.-J.Y, N.P.K.N, K.N.T, and H.-M.S; supervision, I.-J.Y; funding acquisition, I.-J.Y.

CRediT authorship contribution statement

Ly Thi Huong Nguyen: Writing – original draft, Investigation, Formal analysis, Conceptualization. **Nhi Phuc Khanh Nguyen:** Writing – review & editing. **Khoa Nguyen Tran:** Writing – review & editing. **Heung-Mook Shin:** Writing – review & editing. **In-Jun Yang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

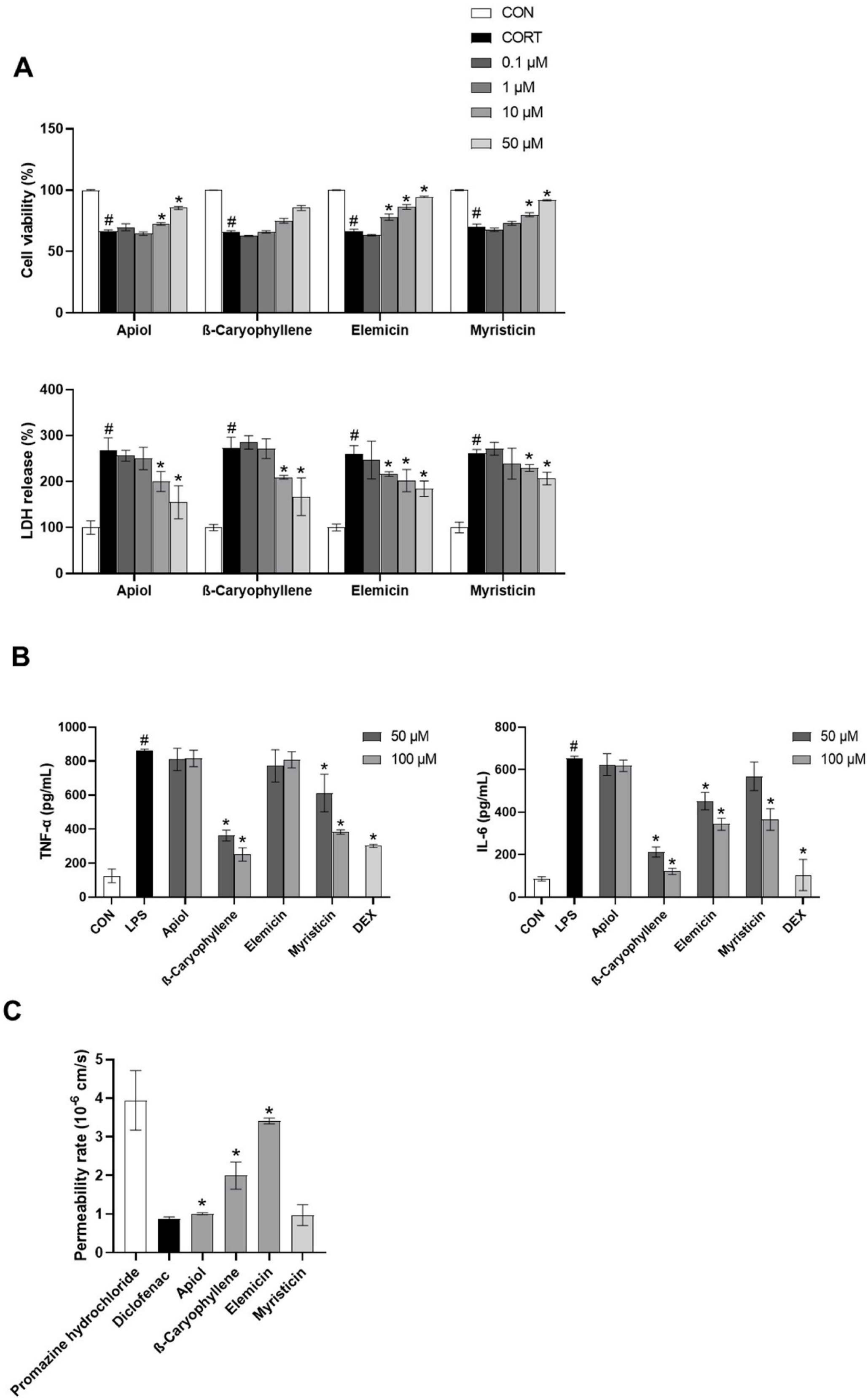

Fig. 10. *In vitro* effect of the PFEO compounds. (A) Their effects on PC12 cells identified using cell viability and LDH release assays. (B) Their effects on BV2 cells identified by measuring the TNF-α and IL-6 levels. Data are presented as the mean ± SEM (n = 3 per group); #p < 0.05 vs. CON; *p < 0.05 vs. CORT or LPS. (C) BBB permeability of the PFEO compounds according to the PAMPA-BBB assay. Data are presented as the mean ± SEM (n = 8 per group), *p < 0.05 vs. diclofenac.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jep.2024.117775.

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